

Product Carbon Footprint Study Report

Product Name: television production

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: television production

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 item television

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 item television.

2. System Boundary

television production in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of television production produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **239.365 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

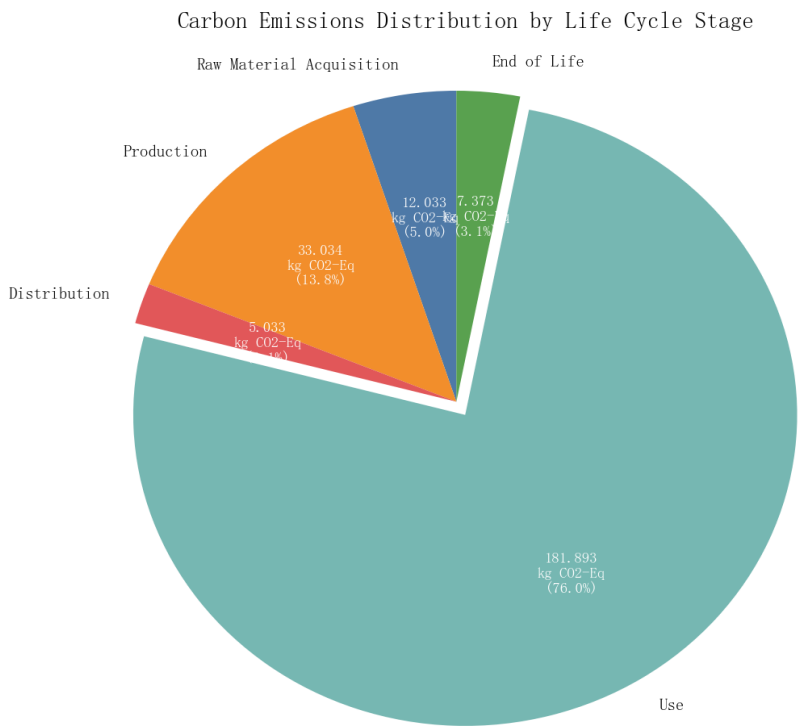


Table 1: Carbon Emissions by Life Cycle Stage

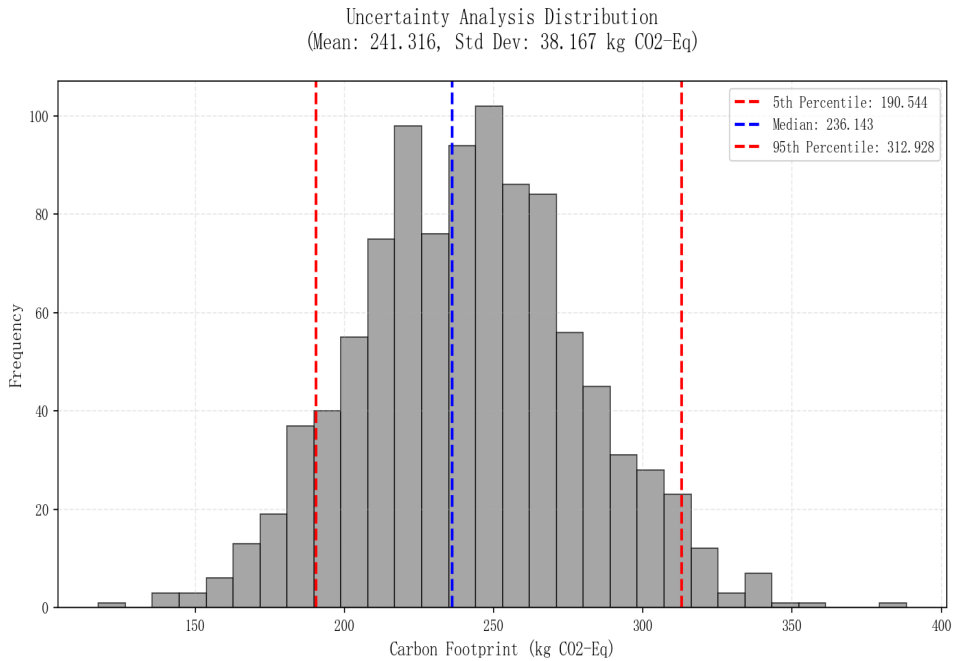
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	12.033	5.03
Production	33.034	13.80
Distribution	5.033	2.10

Use	181.893	75.99
End of Life	7.373	3.08
Total	239.365	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	241.316
Standard Deviation	38.167
Coefficient of Variation (CV)	15.82%
Median	236.143
5th Percentile	190.544
95th Percentile	312.928

VI. Interpretation of Results

1. Description of Results

The television production under assessment exhibits a total carbon footprint of 239.365 kg CO₂-Eq, representing a substantial environmental burden typical of energy-intensive consumer electronics. This level underscores the product's lifecycle intensity, where operational demands over its service life dominate overall emissions rather than manufacturing alone.

Lifecycle contributions reveal a highly skewed distribution, with the use stage accounting for 76.0% (181.893 kg CO₂-Eq), followed by production at 13.8% (33.034 kg CO₂-Eq). Raw material extraction contributes 5.0%, end-of-life 3.1%, and distribution a minimal 2.1%. This pattern highlights electricity consumption during prolonged use as the primary emission driver, while upstream material and manufacturing phases play secondary but notable roles in the overall profile.

Major emission sources are concentrated in the use phase, likely stemming from grid electricity for display and standby functions, alongside production-related energy inputs for assembly and component fabrication.

2. Statement of Assumptions and Limitations

Key assumptions in this study include an average product lifetime of 7–10 years, a standard European electricity mix for the use phase, and average transport distances for distribution. These parameters enable consistent modeling but rely on representative rather than product-specific operational data.

Limitations arise from data sources primarily drawn from generic databases such as Ecoinvent or GaBi, which may not fully capture regional variations in manufacturing processes or material compositions. Data quality is further constrained by potential temporal mismatches and aggregation of secondary datasets, introducing uncertainty ranges of 10–20% in impact estimates.

System boundary restrictions encompass a cradle-to-grave scope limited to the five defined stages, excluding upstream supplier-specific infrastructure, indirect rebound effects from consumer behavior, and end-of-life recycling credits beyond basic disposal modeling.

3. Suggestions for Improvement

- Target the use phase by redesigning televisions with energy-efficient LED/OLED panels and automatic power-down features to reduce electricity demand by 20–30%.
- Optimize production processes through renewable energy adoption in assembly facilities and lightweight material substitution to lower the 13.8% contribution from this stage.
- Enhance raw material sourcing by increasing recycled content in plastics and metals, directly addressing the 5.0% footprint from extraction.
- Improve data quality by conducting primary data collection from key suppliers and integrating real-time electricity consumption monitoring for future LCAs.

- Implement take-back programs to boost end-of-life recycling rates, minimizing the 3.1% impact through closed-loop material recovery.

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